

Aneesh Ganti

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EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science and Mathematics

Expected: May 2027

GPA: 3.70/4.00

Relevant Coursework: • Data Structures • Algorithms • Distributed Systems • Database Systems • Cloud Computing • Software Design • System Programming • Computer Architecture • Machine Learning • Natural Language Processing • Computer Vision • Data Science

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Brunswick Corporation

January 2026 — Present

- Engineered physics-informed propeller recommendation tool to serve **400+** boating users and save **\$25,000** in manual testing costs.
- Developed agentic cryptanalysis research to decode **70+** encrypted CAN signals and find helm steering, throttle, and fuel vulnerability.
- Built progressive web app with TypeScript, Workbox, and Azure to cut test analysis time for on-water engineers from **3 hours** to **<4s**.

Software Engineer | Promo Pigeon (iVenture 12)

February 2026 — Present

- Launched Gemini-powered AI agent to generate product mockups from client emails and cut response time from **2 days** to **60 seconds**.
- Designed spaCy and PyMuPDF document parser to extract and store purchase order data and image assets for **2,500+** daily proofs.
- Architected a Dockerized artwork generation engine on AWS ECR to eliminate a 6-step manual design workflow for labels and flyers.

Software Engineer Intern | Tekweld Manufacturing

May 2025 — February 2026

- Built sales outreach tool with Python and AWS to create and ship personalized samples to **700+** businesses with **14%** conversion rate.
- Accelerated product design from **1 week** to **15 minutes** using an LLM-driven pipeline for label printing and visual quality assurance.
- Shipped React and FastAPI web application on AWS Lightsail to manage order fulfillment and brand assets for **90,000+** client leads.

RESEARCH EXPERIENCE

Undergraduate Researcher | University of Illinois Urbana-Champaign

February 2026 — Present

- Developed self-supervised DINOv3 and SAM2 segmentation model with PyTorch for annotation-free prompted instance segmentation.
- Designed ViT-conditioned diffusion model to predict 3D object trajectories from RGB images for **17-second** robot motion planning.
- Integrated 2D-to-3D lifting on TAPIP3D keypoints to generate motion training labels from video and scale training data by **260%**.

Machine Learning Researcher | University of California, Santa Barbara

June 2023 — August 2023

- Trained [TensorFlow](#) and [Keras](#) DNN to predict battery degradation state and remaining useful life from **10,000+** NASA charge cycles.
- Reduced battery health prediction error by **18%** over BLS-RVM and LSTM baselines to validate generalization across battery models.
- Collaborated with UCSB and MIT AI researchers on EV battery diagnostics methods and presented findings at the capstone seminar.

LEADERSHIP AND ACTIVITIES

Software Engineer | Spaceshot Rocketry Program

August 2024 — Present

- Implemented C++ [Extended Kalman Filter](#) to estimate rocket dynamics with **10-ft** accuracy in **75,000-ft** launch for flight tracking.
- Built telemetry interface with JavaScript to track MQTT transmissions and ensure zero packet loss in turret control during launches.

Software Contributor | ZenGM Sports Management

January 2026 — Present

- Architected a serverless [franchise simulation](#) with **50,000+** active users using multi-threaded Shared Workers, IndexedDB, and React.
- Developed TypeScript pipeline with 4-type conflict detection to replace randomized scheduling with real-world sports data ingestion.

Data Engineer | PhinD Experts (iVenture 10)

May 2025 — August 2025

- Automated CI/CD with GitHub Actions, Docker, and AWS to deploy a **4,000+** profile PhD recruiter-candidate matching platform.
- Revamped Airflow ETL pipeline to scrape LinkedIn and Google Scholar and cut PhD candidate matching time from **1 minute** to **25s**.

Software & Simulations Lead | NASA Human Lander Challenge

August 2024 — July 2025

- Engineered [ML pipeline](#) using Python, ANSYS, and Scikit-learn to improve in-space cryogenic propellant transfer efficiency by **84%**.
- Designed agnostic protocol for NASA's Artemis program and led team to win Best Presentation and **\$1,200** prize over **30+** universities.

SKILLS

Programming Languages: Python, C++, SQL, TypeScript, Java, JavaScript, HTML, CSS, Swift, R, Verilog, MIPS

Full-Stack & Backend: FastAPI, PostgreSQL, REST APIs, React, Node.js, Flask, MongoDB, Express, Redis, WebSockets

GenAI Tools: OpenAI API, Gemini, LangChain, ONNX, LlamaIndex, Claude Code, RAG, Hugging Face, Pinecone, vLLM, PEFT, LoRA

ML & Data Science: PyTorch, Scikit-learn, TensorFlow, NumPy, Pandas, OpenCV, CUDA, spaCy, SciPy, Matplotlib, MLflow

Cloud & Infrastructure: AWS, Docker, CI/CD, Kubernetes, Git, GitHub, GitHub Actions, Google Cloud, Linux, SLURM, Jest, Agile